Pasadena. (Baade was the man who seventeen years earlier, with Zwicky, had identified supernovae and proposed that neutron stars power them—Chapter 5.) The Carnegie Institution owned the 2.5-meter (100-inch) optical telescope on Mount Wilson, until recently the world's largest; Caltech, down the street in Pasadena, had just finished building the larger 5-meter (200-inch) telescope on Palomar Mountain; and the Carnegie and Caltech astronomers shared their telescopes with each other. At his next scheduled observing session on the Palomar 5-meter (Figure 9.3a), Baade photographed the error box on the sky where Smith said Cyg A lies. (This spot on the sky, like most spots, had never before been examined through a large optical telescope.) When Baade developed the photograph, he could hardly believe his eyes. There, in the error box, was an object unlike any ever before seen. It appeared to be two galaxies colliding with each other (center of Figure 9.3d). (We now know, thanks to observations with infrared telescopes in the 1980s, that the galaxy collision was an optical illusion. Cyg A is actually a single galaxy with a band of dust running across its face. The dust absorbs light in just such a way as to make the single galaxy look like two galaxies in collision.) The whole system, central galaxy plus radio source, would later come to be called a *radio galaxy.*

Astronomers were convinced for two years that the radio waves were being produced by a galactic collision. Then, in 1953, came another surprise. R. C. Jennison and M. K. Das Gupta of Lovell's Jodrell Bank team studied Cyg A using a new interferometer consisting of two telescopes, one fixed to the ground and the other moving around the countryside on a truck so as to cover, one after another, a number of "spots" on the "bowl" of an imaginary 4-kilometer-square telescope (see left half of Figure 9.2). With this new interferometer (Figures 9.3b, c), they discovered that the Cyg A radio waves were not coming from the "colliding galaxies," but rather from two giant, roughly rectangular regions of space, about 200,000 light-years in size and 200,000 light-years apart, on opposite sides of the "colliding galaxies." These radio-emitting regions, or *lobes* as they are called, are shown as rectangles in Figure 9.3d, together with Baade's optical photograph of the "colliding galaxies." Also shown in the figure is a more detailed map of the lobes' radio emission, constructed sixteen years later using more sophisticated interferometers; this map is shown as thin lined contours that exhibit the brightness of the radio emission in the same way as the contours of a topographic map exhibit the height of the land. These